CSE276 — ARTIFICIAL INTELLIGENCE FOUNDATIONS · WEEK 6 · UNIT III

Intelligent Agents & the PEAS Framework

How AI systems perceive, decide, and act

Lecture 11

What do these systems have in common?



Robot Vacuum



Self-Driving Car



Smart Thermostat

They all: Observe → Decide → Act

Learning Objectives

- Explain what an intelligent agent is.
- Explain the relationship between an agent and its environment.
- Understand sensors and actuators.
- Explain percepts and actions.
- Understand the agent-environment interaction loop.
- Explain the basic idea of rational behavior.
- Identify agents in real-world systems.
- Use the PEAS framework to describe an AI task.
- Construct PEAS descriptions for simple AI systems.

What Is an Agent?

An agent is something that **perceives** its environment and **takes actions**.



A Simple Real-World Analogy

Humans are excellent examples of agents. We interact with the physical world continuously.

Sensors (Perceive)

Eyes, ears, skin receive information.



Decision Process

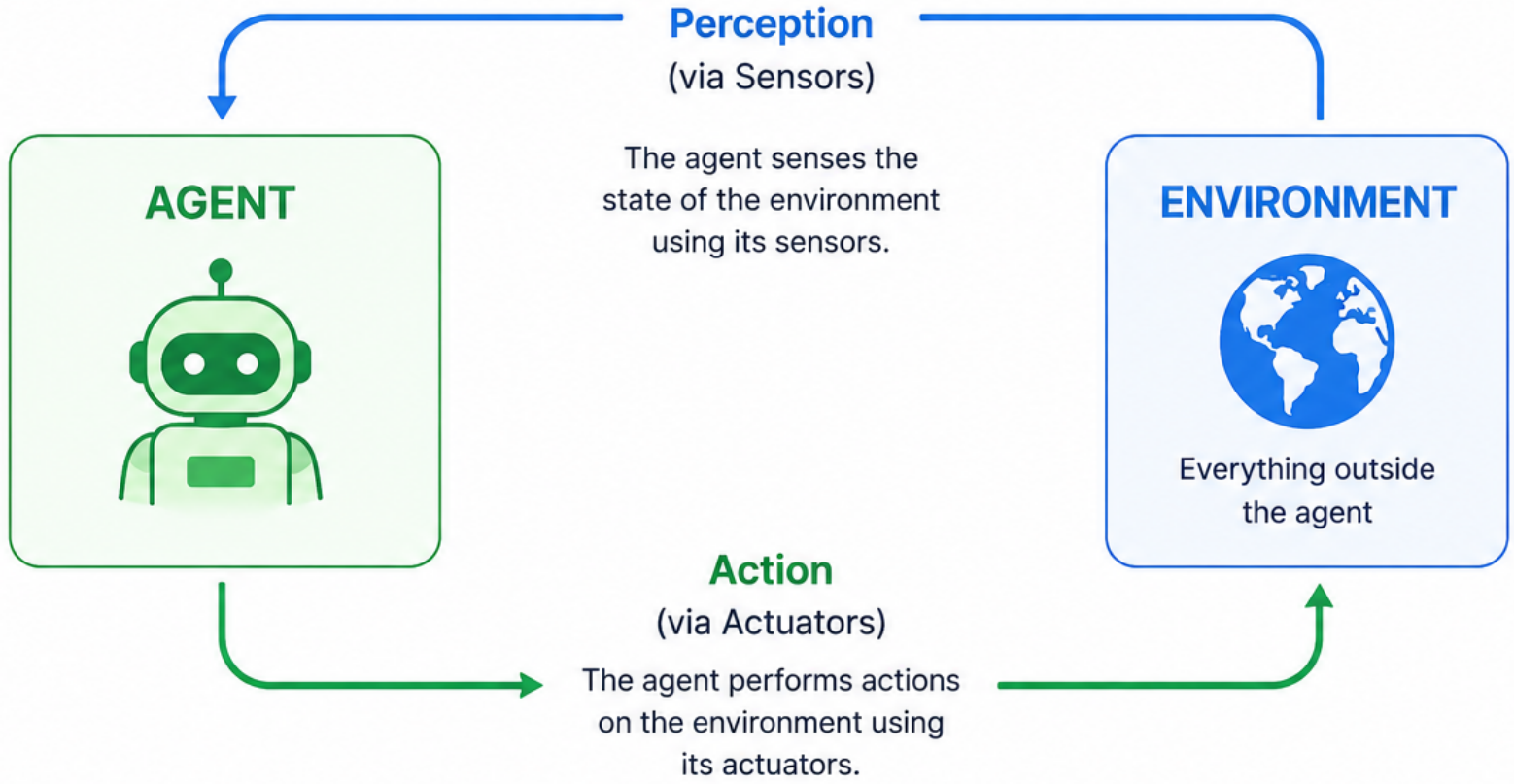
The Brain processes info and makes choices.

Actuators (Act)

Hands, legs, voice affect the world.

Agent–Environment Interaction

An agent perceives its environment through sensors and acts upon it through actuators.



This forms a continuous loop:

Perceive → Think/Decide → Act → Environment Changes → Perceive again

Sensors and Actuators



Sensors

Things through which an agent **receives information**.

- Camera
- Microphone
- GPS
- Keyboard input



Actuators

Things through which an agent **affects the environment**.

- Motor / Wheels
- Robotic arm
- Speaker
- Display screen

CONCEPT DEFINITION

What is Perception?

A **percept** is the information an agent receives from its environment through its sensors at a particular moment.



Examples:

Robot Vacuum

Camera sees an obstacle, bump sensor detects a wall, battery sensor reports low power.

Self-Driving Car

Camera sees a pedestrian, GPS provides location coordinates, Radar detects another vehicle.

CONCEPT DEFINITION

What Is an Action?

An **action** is what an agent does to affect or change its environment.

Examples:

Robot Vacuum

Turn left, move forward, stop, return to charging station.

Navigation System

Recommend a route, recalculate path, output voice warning.



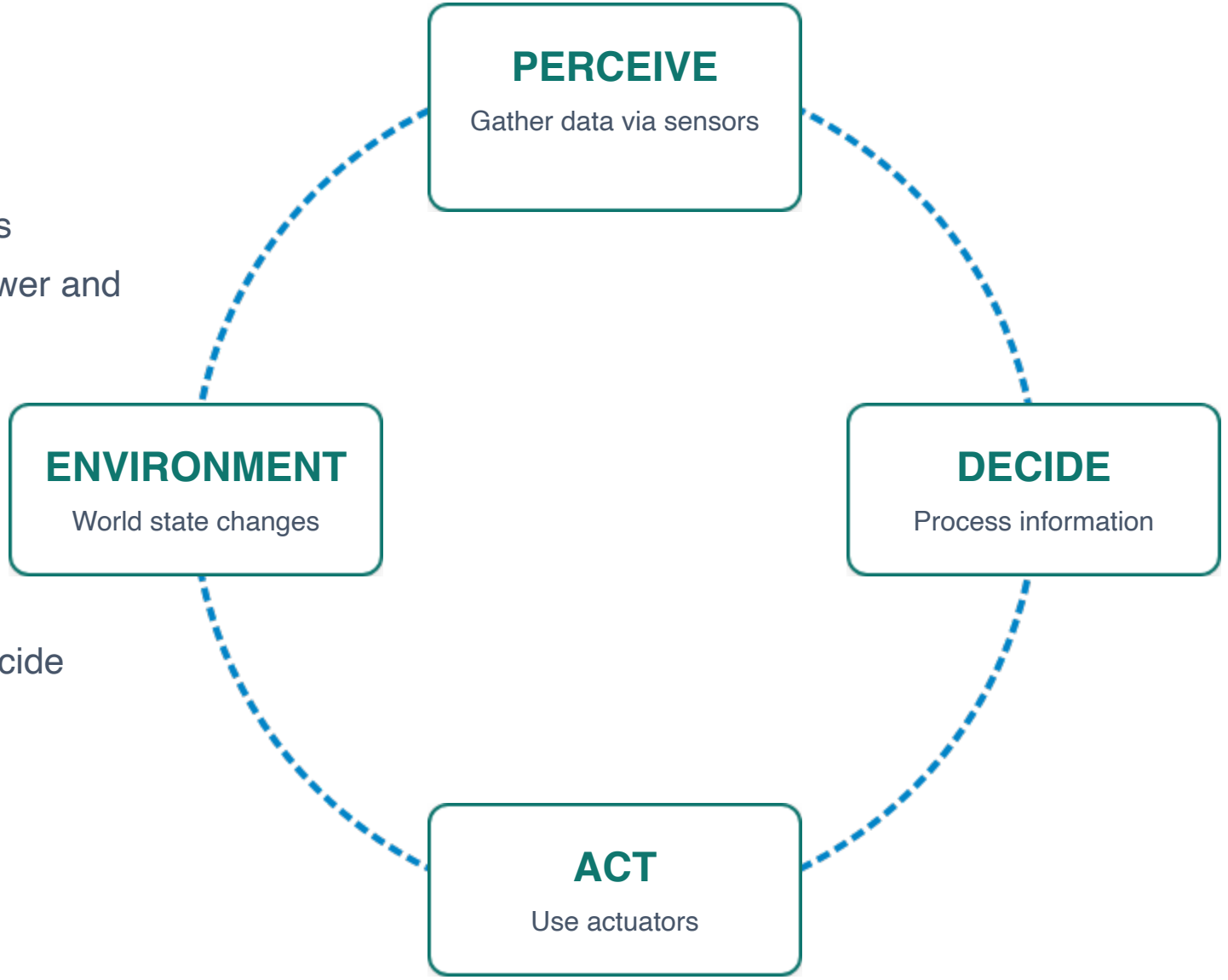
The Agent Loop

An intelligent agent **continuously interacts** with its environment rather than simply producing one answer and stopping.

The environment is dynamic.

Actions change the environment.

The agent must perceive the *new* state to decide what to do next.



DISCUSSION

Is Every Program an Agent?



Calculator App

Input: 2 + 2 → Output: 4

It receives input and produces output, but it does not naturally fit the richer **continuous interaction loop** with an external environment.



Robot Vacuum

Perceives → Chooses Action → Changes World

In AI, we reserve the term "agent" for systems that exhibit this continuous, autonomous interaction with a task environment.

Goal-Oriented Behavior

Agents usually choose their actions based on an objective they are trying to achieve.

Robot Vacuum

Goal: Clean the floor efficiently without getting stuck.

Navigation System

Goal: Reach the destination safely in the shortest time.

Chess AI

Goal: Checkmate the opponent's king (win the game).

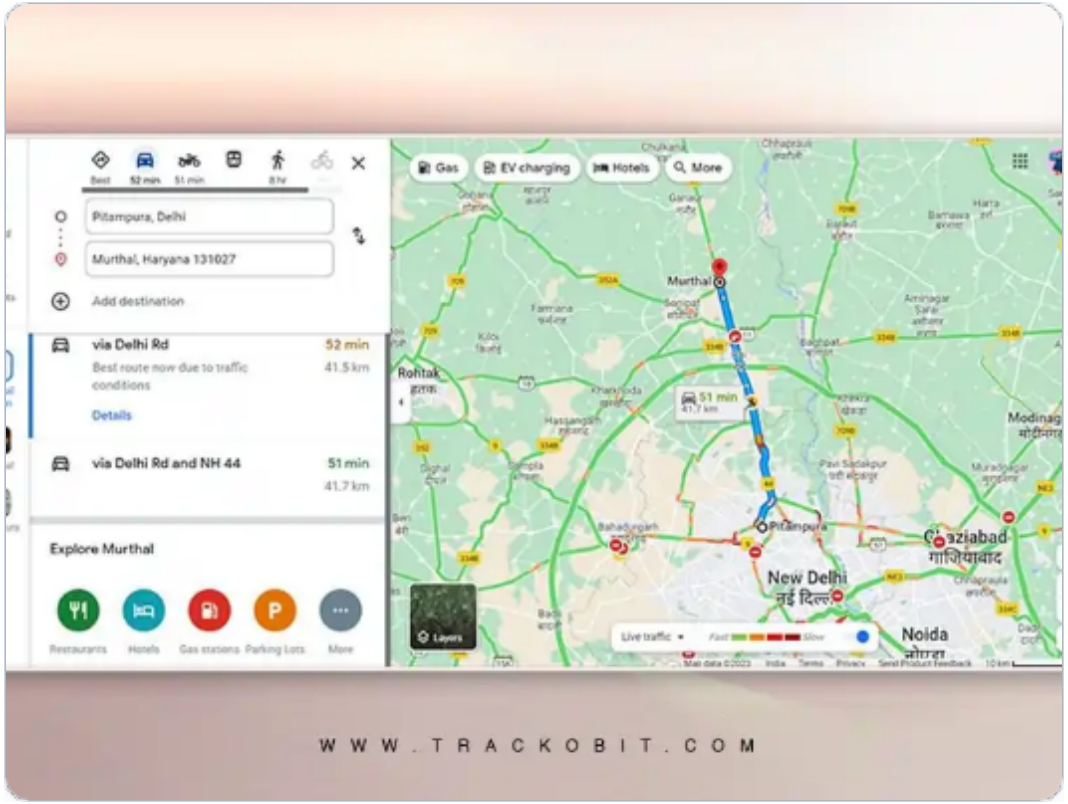
Smart Thermostat

Goal: Maintain desired temperature while minimizing energy use.

Rational Agent: Intuition

What makes an agent "intelligent" or "good"?

> A **rational agent** chooses actions that are expected to achieve the best outcome according to its goals and available information.



Example: Navigation Agent

Route A: 30 minutes

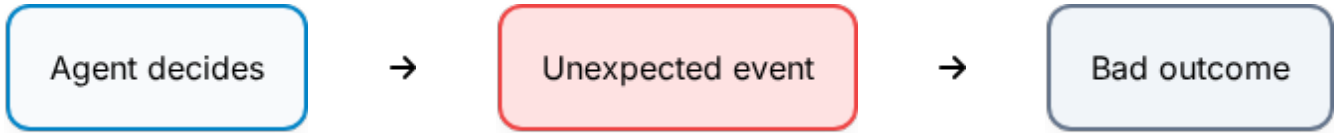
Route B: 60 minutes

If both are safe and available, choosing Route A is the **rational choice** to maximize performance.

COMMON MISCONCEPTION

Rational ≠ Always Correct

A rational decision can still produce a bad outcome. Was the agent irrational?



Why Rationality is not Omniscience:

Incomplete Information

The agent can't see around corners.

Uncertainty

Sensors might be noisy or imprecise.

Unexpected Events

The environment behaves unpredictably.

One Concept, Many Applications

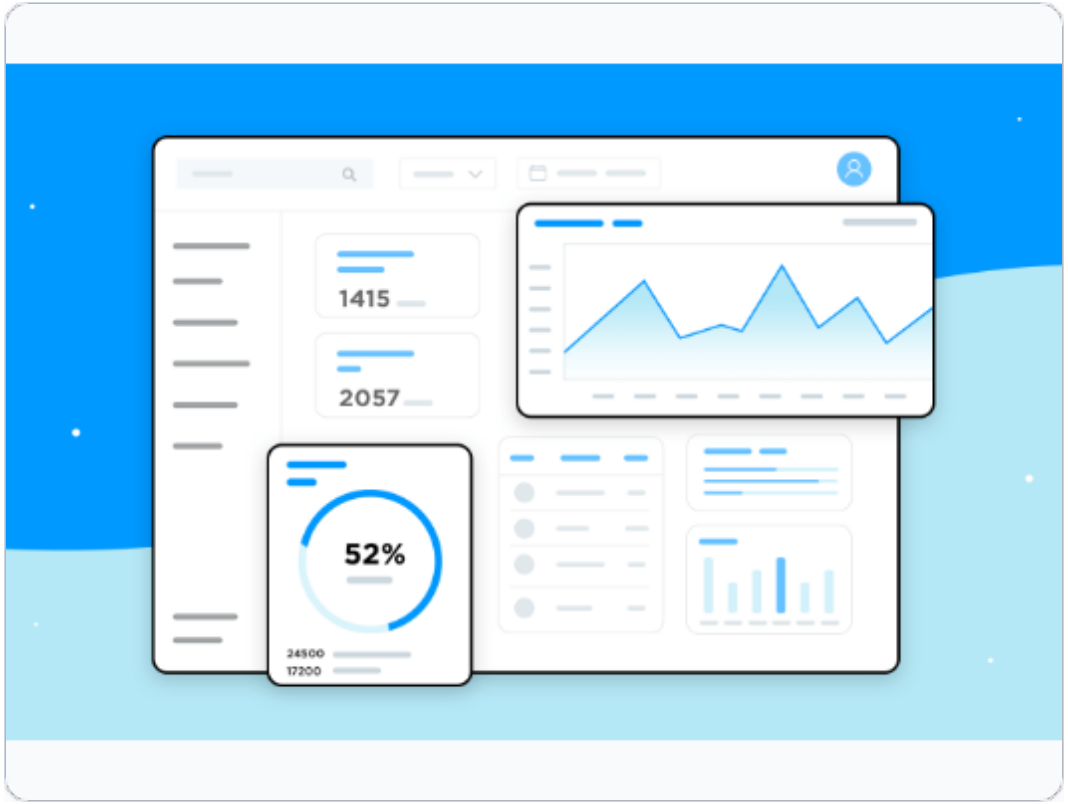
Agent Type	What it Perceives (Sensors)	What it Acts upon (Actuators)
Robot Vacuum	Room, obstacles, dirt, battery level	Moves wheels, turns on brushes
Self-Driving Car	Roads, vehicles, pedestrians, signs	Steers, brakes, accelerates
Chess AI	Board state, opponent's moves	Selects and moves a piece
Recommendation System	User clicks, watch history, likes	Displays recommended content

KEY INSIGHT

Software Agents (Softbots)

An agent does not need a physical body.

Agents can exist entirely in digital environments, perceiving data streams and acting via digital commands.



Examples:

- Trading or Budgeting Assistants
- E-commerce Recommendation Systems
- Video Game AI
- Chatbots and AI Assistants

Why Do We Need PEAS?

Suppose someone gives you this project assignment:

"Build an AI agent for driving."

That is entirely too vague!

- What counts as success?
- Where does it operate?
- What can it observe?
- What can it do?

What Is PEAS?

PEAS is a framework used to specify the task environment of an intelligent agent before we begin designing it.

P

Performance Measure

How we evaluate success.

E

Environment

Where the agent operates.

A

Actuators

How the agent acts.

S

Sensors

How the agent perceives.

P: Performance Measure

The **Performance Measure** tells us how we judge whether the agent is doing well.

> *Important distinction: It is about how we (the designers) evaluate success, not simply what the agent "wants."*

Examples for a Self-Driving Car:

- Safety (no accidents)
- Travel time (fast route)
- Comfort (smooth driving)
- Traffic-rule compliance
- Energy efficiency

E: Environment

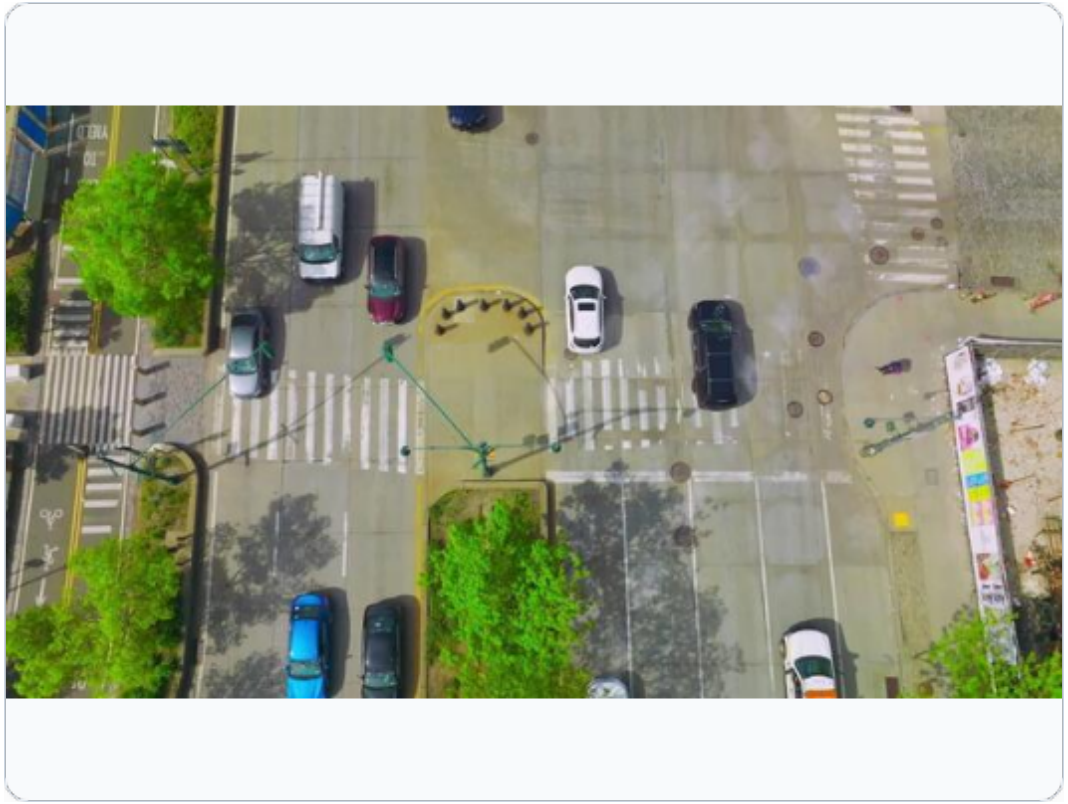
The **Environment** is everything outside the agent that the agent interacts with or that affects its task.

Self-Driving Car Environment:

Roads, other vehicles, pedestrians, traffic lights, weather conditions, road signs.

Robot Vacuum Environment:

Rooms, furniture, walls, dirt, people, pets.



A: Actuators

Actuators are the mechanisms through which an agent takes actions.

Self-Driving Car

- Steering wheel
- Accelerator
- Brakes
- Indicators / Horn

Physical Robot

- Wheels / Tracks
- Robotic arm
- Gripper mechanism

Software Agent

- API calls
- Sending messages
- Executing commands
- Updating a UI

S: Sensors

Sensors provide the agent with information about its environment.

Self-Driving Car

- Cameras
- Radar & LiDAR
- GPS
- Speedometers

Robot Vacuum

- Camera / Optical
- Bump sensors
- Cliff/distance sensors
- Battery gauge

Software Agent

- Keyboard/Mouse input
- Network packets
- Database queries
- Web scraping data

PEAS Example: Self-Driving Car

P Performance Measure	Safety, travel time, comfort, legal compliance, energy efficiency
E Environment	Roads, traffic, pedestrians, weather, traffic signals
A Actuators	Steering, brakes, accelerator, indicators
S Sensors	Cameras, GPS, radar, LiDAR, speed sensors

Class Question: Can you think of one more sensor? One more actuator?

PEAS Example: Robot Vacuum

P Performance Measure	Cleaning quality, time taken, battery efficiency, safety (no stucks)
E Environment	Rooms, furniture, walls, people, pets, floor types
A Actuators	Wheels, brushes, suction mechanism, speaker (beeps)
S Sensors	Cameras, bump sensors, distance/cliff sensors, battery sensor

Class Question: What would happen if the robot had NO sensors?

Final Recap + Exit Question

Intelligent Agent

Perceives → Decides → Acts

Sensors & Actuators

How the agent perceives / How the agent acts

PEAS Framework

Performance Measure, Environment, Actuators, Sensors

Exit Question

Design the PEAS description for a **Chess-Playing AI**.

(Take 2 minutes to think through P, E, A, and S)

Next Lecture:

Rational Agents & Environment Properties

Image Sources



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